STPM PHYSICS · SEMESTER 1 · CHAPTER 7

Rotation of Rigid Bodies

7.1 Torque and Angular Acceleration

DEFINITIONS & LAWS

- **Torque (moment of a force)** about an axis is the product of the force and the perpendicular distance from the axis to the line of action of the force.
- **Couple** is a pair of equal and opposite parallel forces whose lines of action do not coincide; it produces rotation without translation, and its torque is the product of one force and the perpendicular distance between the two forces.
- **Newton's second law of rotation**: the angular acceleration of a rigid body about a fixed axis is directly proportional to the resultant torque and inversely proportional to its moment of inertia.
- Torque is a vector (clockwise/anticlockwise sense must be stated); a force whose line of action passes through the axis has zero torque.
- **Angular acceleration** α is the rate of change of angular velocity, $\alpha = \frac{d\omega}{dt}$ (rad s^{-2}).

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Torque of a force	$\tau = Fr\sin\theta = F_{\perp}r = Fd$	τ = torque (N m); F = force (N); r = distance from axis (m); θ = angle between F and r; $F_{\perp}$ = component of F perpendicular to r; d = r sinθ = perpendicular distance (m)
Couple	$\tau = Fd$	τ = torque of the couple (N m); F = magnitude of each force (N); d = perpendicular separation of the two forces (m)
Torque and angular acceleration	$\tau = I\alpha$	τ = torque (N m); I = moment of inertia (kg m^2); α = angular acceleration (rad s^{-2})
Linear–rotational links	$s = r\theta$; $v = r\omega$; $a_t = r\alpha$	s = arc length (m); v = tangential speed (m s^{-1}); a_t = tangential acceleration (m s^{-2}); θ = angle (rad); ω = angular velocity (rad s^{-1}); α = angular acceleration (rad s^{-2}); r = radius (m)
Equations of rotational motion	$\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; $\omega^2 = \omega_0^2 + 2\alpha\theta$	ω_0, ω = initial and final angular velocities (rad s^{-1}); θ = angular displacement (rad); t = time (s); constant α (rad s^{-2})
Wheel or pulley with a hanging mass	$mg - T = ma$; $Tr = I\alpha$; $a = r\alpha$	m = hanging mass (kg); T = tension (N); a = linear acceleration of mass (m s^{-2}); r = radius of wheel (m); I = moment of inertia (kg m^2); α = angular acceleration (rad s^{-2})

HOW TO ANSWER (MPM STYLE)

- Draw the force and the axis, mark the perpendicular distance d, then write $\tau = F\,d$; state clockwise or anticlockwise.
- For $\tau = I\alpha$ problems, find the net torque about the axis (sum of torques with sign), then $\alpha = \frac{\tau}{I}$; add a linear equation for each hanging mass.
- Convert cm → m and rev min^{-1} → rad s^{-1} before substituting; give τ in N m and α in rad s^{-2}.
- For a couple, state that the resultant force is zero but the resultant torque is Fd.

SECTION A TRAPS

Trap: Using the distance along the rod without sinθ; the perpendicular distance is r sinθ (or use F sinθ). If F is perpendicular to the rod, sinθ = 1.

Trap: Torque is not measured in joules for this subtopic; unit is N m (do not mistake it for energy).

Trap: In pulley problems, using T = mg for the string tension; the hanging mass accelerates, so T = m(g − a).

Trap: Using F = ma for rotation without I: a body with larger I has smaller α for the same torque.

TRICKS

Trick: Limiting case: if the line of action passes through the pivot, d = 0 and τ = 0.

Trick: Ratio method: for the same τ, $\alpha \propto \frac{1}{I}$; doubling the radius of gyration ($I \propto r^2$) quarters α.

7.2 Moment of Inertia

DEFINITIONS & LAWS

- **Moment of inertia** I of a body about an axis is the sum of the products of the mass of each particle and the square of its perpendicular distance from the axis, $I = \Sigma mr^2$; it is the measure of the body's resistance to angular acceleration (rotational inertia).
- Moment of inertia depends on the mass, on how the mass is distributed about the axis, and on the position of the axis; the same body has different I about different axes.
- Moment of inertia is the rotational analogue of mass: $\tau = I\alpha$ corresponds to $F = ma$.
- A body with its mass distributed further from the axis has a larger I and so is harder to start or stop rotating.
- A flywheel has most of its mass at the rim to give a large I, so that it resists changes in angular speed (stores rotational kinetic energy).

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Point mass	$I = mr^2$	I = moment of inertia (kg m^2); m = mass (kg); r = perpendicular distance from the axis (m)
System of masses	$I = \Sigma m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + \ldots$	I = moment of inertia (kg m^2); m_i = mass of the i-th particle (kg); r_i = its perpendicular distance from the axis (m)
Thin hoop or ring	$I = MR^2$	M = mass (kg); R = radius (m); about its axis
Solid disc or cylinder	$I = \frac{1}{2}MR^2$	M = mass (kg); R = radius (m); about its axis
Solid sphere	$I = \frac{2}{5}MR^2$	M = mass (kg); R = radius (m); about a diameter
Thin spherical shell	$I = \frac{2}{3}MR^2$	M = mass (kg); R = radius (m)
Thin rod about its centre	$I = \frac{1}{12}ML^2$	M = mass (kg); L = length (m); axis through the centre
Thin rod about one end	$I = \frac{1}{3}ML^2$	M = mass (kg); L = length (m); axis through one end
Radius of gyration	$I = Mk^2$	k = radius of gyration (m); M = mass (kg); I = moment of inertia (kg m^2)

HOW TO ANSWER (MPM STYLE)

- Identify the axis first; calculate the distance r of each mass from that axis (perpendicular distance), square it, multiply by the mass, and add.
- For a composite body, add the individual I values about the same axis; show every term with units kg m^2.
- For comparing bodies of the same mass and radius, give the order by mass distribution: hoop > disc > solid sphere ($\frac{I}{MR^2}$ = 1, 0.5, 0.4).
- Link to $\tau = I\alpha$ when asked for the angular acceleration of the body.

SECTION A TRAPS

Trap: The distance in Σmr^2 is measured from the axis, not from the centre of the body or from the other mass; a mass lying on the axis contributes zero.

Trap: Doubling the distance of a mass from the axis multiplies its contribution by 4 (r^2), not by 2.

Trap: Using the formula for the wrong axis: rod about the centre is $\frac{ML^2}{12}$, about the end is $\frac{ML^2}{3}$ (four times larger); disc about its own axis is $\frac{1}{2}MR^2$.

Trap: Moment of inertia of a rigid body is not a fixed property of the body; it is different about different axes and changes if the mass distribution is changed.

TRICKS

Trick: Keep the order I(hoop) > I(disc) > I(solid sphere) for the same M and R, since less mass at the rim lowers I.

Trick: Unit check: I has unit kg m^2; a result in kg m^{-2} or kg m indicates a missing or extra power of r.

7.3 Angular Momentum and Rotational Kinetic Energy

DEFINITIONS & LAWS

- **Angular momentum** L of a rigid body about an axis is the product of its moment of inertia and its angular velocity, $L = I\omega$ (kg m^2 s^{-1}); for a point mass $L = mvr$.
- **Principle of conservation of angular momentum**: the total angular momentum of a system remains constant if no resultant external torque acts on the system.
- **Rotational kinetic energy** is the energy a rotating body possesses because of its angular velocity, $\frac{1}{2}I\omega^2$.
- Newton's second law in terms of angular momentum: the resultant torque is the rate of change of angular momentum, $\tau = \frac{dL}{dt}$.
- When the moment of inertia of a system decreases (e.g. a skater draws in the arms), ω increases so that L is constant; the rotational kinetic energy increases because work is done by the skater (internal forces).
- In a collision or coupling of rotating bodies about the same axis (inelastic), angular momentum is conserved but kinetic energy is lost.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Angular momentum	$L = I\omega = mvr$	L = angular momentum (kg m^2 s^{-1}); I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); mvr applies to a point mass in circular motion
Torque and angular momentum	$\tau = \frac{dL}{dt} = I\frac{d\omega}{dt} = I\alpha$	τ = torque (N m); α = angular acceleration (rad s^{-2}); I constant
Conservation of angular momentum	$I_1\omega_1 = I_2\omega_2$	I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); no external torque
Coupling of two rotating bodies	$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$	I_1, I_2 = moments of inertia (kg m^2); ω_1, ω_2 = initial angular velocities (rad s^{-1}); ω = common final angular velocity (rad s^{-1})

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Rotational kinetic energy	$K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$	K = rotational kinetic energy (J); I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); L = angular momentum (kg m^2 s^{-1})
Rolling without slipping	$v = r\omega$; $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$	v = speed of centre (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1}); K = total kinetic energy (J); m = mass (kg); I = moment of inertia (kg m^2)
Rolling down an incline	$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$	h = height of incline (m); starts from rest; m = mass (kg); v = speed at the bottom (m s^{-1}); ω = angular velocity (rad s^{-1})
Solid cylinder rolling down an incline	$v = \sqrt{\frac{4gh}{3}}$	v = speed at the bottom (m s^{-1}); h = height (m); g = gravitational acceleration (m s^{-2}); from rest
Work done by a torque	$W = \tau\theta$	W = work done (J); τ = torque (N m); θ = angle turned (rad)
Power of a torque	$P = \tau\omega$	P = power (W); τ = torque (N m); ω = angular velocity (rad s^{-1})

HOW TO ANSWER (MPM STYLE)

- State the principle of conservation of angular momentum with the condition "no resultant external torque" before using $I_1\omega_1 = I_2\omega_2$.
- Find I for each stage ($I = \Sigma mr^2$), substitute with ω in rad s^{-1}, and give the final ω with unit rad s^{-1} (convert to rev min^{-1} only if asked).
- When asked for the change in kinetic energy, calculate $\frac{1}{2}I\omega^2$ before and after and state the source (work done by the person/internal forces) or the loss (inelastic).
- For rolling, include both translational and rotational kinetic energy; state $v = r\omega$ as the no-slip condition.

SECTION A TRAPS

Trap: Only L is conserved when I changes; $\frac{1}{2}I\omega^2$ is not conserved. If I decreases by half, ω doubles and K doubles ($K = \frac{L^2}{2I}$).

Trap: Angular momentum is conserved only with zero resultant external torque; do not claim it for a body acted on by a net torque.

Trap: Rolling: total kinetic energy is greater than $\frac{1}{2}mv^2$; a body with a smaller $\frac{I}{MR^2}$ (solid sphere) reaches the bottom of the incline first and has the larger v, independent of m and R.

Trap: Forgetting to convert rev min^{-1} to rad s^{-1} before using $\frac{1}{2}I\omega^2$; this gives an error of a factor $(\frac{60}{2\pi})^2$.

TRICKS

Trick: Skater: $I_2 = \frac{I_1}{n}$ gives $\omega_2 = n\omega_1$ and $K_2 = nK_1$; use the ratio directly without the numbers.

Trick: Unit check: $\frac{1}{2}I\omega^2$ gives kg m^2 s^{-2} = J; if the unit is not J, ω has not been in rad s^{-1}.